

The mechanisms beneath the urban buffer against depression

Junyi Li ^{a,b}, Tian Gan ^{a,c}, Shenghan Lyu ^a, Hongying Wu ^a, Simone Macrì ^d,
Maurizio Porfiri ^{a,b,c,e}*

^a Center for Urban Science and Progress, Tandon School of Engineering, New York University, 370 Jay Street, Brooklyn, 11201, NY, USA

^b Department of Civil, Urban and Environmental Engineering, Tandon School of Engineering, New York University, 6 MetroTech Center Jacobs Hall, RH 400, Brooklyn, 11201, NY, USA

^c Department of Mechanical and Aerospace Engineering, Tandon School of Engineering, New York University, 5 MetroTech Center, Brooklyn, 11201, NY, USA

^d Centre for Behavioural Sciences and Mental Health, Istituto Superiore di Sanità, 299 Viale Regina Elena, Rome, 00161, Italy

^e Department of Biomedical Engineering, Tandon School of Engineering, New York University, 433 First Avenue, 10010, NY, USA

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ABSTRACT

Living in a city brings a broad range of risks and protective factors to urban dwellers, yet how these factors interact within cities to impact their mental health remains unclear. Here, we integrate urban scaling and causal discovery to investigate how depression prevalence varies with the population size, and how urban determinants shape this relationship across 826 cities in the United States. Accounting for spatial dependence between cities, we find that depressive disorder scales sublinearly with population size, supporting the view that it is less likely to experience depression in more populous cities. The observed sublinear growth is accompanied by superlinear growth in several candidate determinants (including excessive drinking, single parenthood, and access to mental health providers), alongside sublinear growth in physical disability and street intersection density. Using causal discovery on population-adjusted data, we identify physical disability and excessive drinking as risk factors for depression, in contrast to economic prosperity and walkability that emerge as protective factors against depression. These findings highlight cities as systems intertwined with risks and protections, offering new insights that could inform system-oriented strategies to promote psychological well-being.

1. Introduction

Mental health disorders are a dire concern in global health (Mao et al., 2025). According to the Global Burden of Disease Consortium, their incidence increased by 15.23% between 1990 and 2021 (Mao et al., 2025). Depression is among the most widespread mental disorders, with a prevalence rate of 9.2% in the United States (U.S.) in 2020 (Goodwin et al., 2022) and 6.5% across 30 European countries in 2018–2022 (de la Torre et al., 2023).

Depression arises from a complex interplay of biological, psychological, social, and environmental factors (Marx et al., 2023; Sullivan et al., 2000). The seminal work by Caspi et al. (2003) showed that a specific polymorphism in the serotonin transporter gene modulates individual reactivity towards major stressful life events and the onset of depression thereof. This work incited numerous studies investigating whether other variations in gene function adjust individual susceptibility to depression. Genome-wide association studies identified specific gene variants associated with higher vulnerability to major depressive

disorder (Howard et al., 2019). Consistent with the multifactorial etiology of depression, numerous genetic variants (with larger and smaller effect sizes) have been identified. Polygenic risk scores aggregate the contribution of these variants into a single measure, allowing the stratification of individual subtypes as a function of their vulnerability to depression (Uffelmann et al., 2021). While genetic predispositions may modulate individual vulnerability to depressive disorders (Shi et al., 2025), environmental stressors and protective factors may also influence the likelihood of their occurrence (Arnaud-Soler et al., 2019). Major life events such as job loss and family disruptions have been associated with high depression risk in previous studies (Fagan & Churchill, 2012; Riumallo-Herl et al., 2014).

Living in cities adds another layer of complexity to this multifactorial picture (J. Xu et al., 2023). Understanding how specific urban features modulate disease onset and course may foster a novel perspective in patient care. Specifically, while the focus on environmental stressors greatly benefits personalized care (for example, promoting individual stress management), understanding the role of urban characteristics

* Corresponding author.

E-mail address: mporfiri@nyu.edu (M. Porfiri).

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may enable population-level interventions (for example, modifying those city attributes that increase vulnerability to disease). Despite this innovation potential, the extent of urban influence on depression is to be fully detailed, with contradictory findings about the benefits and drawbacks of living in a city. On one hand, adult individuals in urban environments have been reported to have a greater risk of experiencing depression (Sundquist et al., 2004). Proximity to urban areas has been associated with higher depression rates compared to rural areas (C. Xu et al., 2023). On the other hand, recent findings have challenged this view, whereby Breslau et al. (2014) reported that depression prevalence was slightly lower in large metropolitan areas compared to smaller cities. Along a similar line, Stier et al. (2021) pointed at a lower depression rate in larger cities, potentially reflecting that denser socioeconomic networks may bring a buffer against depression.

These evidently contradictory findings underscore the need for a systematic, data-driven framework to disentangle the relationship between city living and depression. City residents are exposed to a broad spectrum of risks for major depressive disorder, spanning environmental, social, and economic factors. Environmental factors, such as traffic noise (Jacobs et al., 2024), air pollution (Nobile et al., 2023), and low walkability (Liu & Liao, 2025), may exacerbate stress levels and heighten the risk of depression. In highly connected urban environments, social contagion of unhealthy behaviors, such as drug overdose (Mullachery et al., 2022) and excessive drinking (Weitzman et al., 2003), may further exacerbate the risk of depression (Bartoli et al., 2014; Boden & Fergusson, 2011). Likewise, economic insecurity and the high expense of living may introduce financial pressure and constrain access to healthcare (Greenberg et al., 2023, 2021). At the same time, cities can provide protective resources, such as greater access to mental health professionals (Gan et al., 2025), opportunities for physical activity (Pearce et al., 2022), and shorter distances to green space (Li et al., 2026), which foster resilience and mitigate vulnerability to major depressive disorder. Contradictory observations about depression prevalence in cities, along with the coexistence of these opposing urban forces, underscore the complex and context-dependent nature of how urbanization affects depression prevalence. We propose the application of urban scaling (Bettencourt, 2013) to identify whether specific urban features, such as health behaviors, family structure, economic performance, walkability of built environment, and presence of mental health resources, act as moderators of depression prevalence across cities of different sizes.

Urban scaling offers a powerful lens for understanding complex health outcomes shaped by diverse urban contexts (McCulley et al., 2022). For example, analysis in Brazil shows how socioeconomic, infrastructural, and service variables follow superlinear scaling patterns, implying per capita values increase with population size, and that deviations are often associated with treatment and policy interventions on mental health (Meirelles et al., 2018). Cross-national evidence indicates diverse patterns across nations: whereas large cities in the U.S. and Sweden tend to foster healthier lifestyles and lower obesity (Gan et al., 2025; Rocha et al., 2015), cities in the United Kingdom and Brazil are associated with higher smoking rates, excessive drinking, high risks of crime and distress (Puttock et al., 2025; Rocha et al., 2015). Healthcare facilities also scale differently, with primary care clinics in South Korea scaling superlinearly, and hospitals slightly sublinearly (Um et al., 2009). The complex, context-dependent scaling of health-related factors underscores the need to investigate the urban determinants of mental health. Here, we apply urban scaling to study how depression prevalence varies with respect to the city population. In addition, we investigate scaling patterns of several variables (or their most reasonable proxies) that have already been implicated in depression, and for which population-level mitigation strategies can be implemented. As potential moderators of depression prevalence, we focus on excessive drinking (as a proxy for alcohol use disorder although not a direct measure) (McHugh & Weiss, 2019), physical disabilities (Noh et al., 2016), single-parent households (as a proxy

for marital dissolution) (Sbarra et al., 2014), access to mental health resources (Goldman et al., 2001), street intersection density (as a proxy for walkability) (Lee & Sener, 2016), and gross domestic product (as a proxy for economic performance) (Arias et al., 2022).

Urban scaling is also apt to identify potentially causal associations among urban factors, moving beyond its traditionally descriptive role (Gan et al., 2025; Succar & Porfiri, 2024). To this end, one employs the deviation between each city's observed outcome and the value predicted by urban scaling laws to infer a directed acyclic graph (DAG) among the selected features, thereby uncovering the underlying causal associations rather than relying solely on correlations. Specifically, we examine the relationship between depression and urban conditions across 826 U.S. cities, incorporating behavioral health dimensions (for example, physical disabilities and excessive drinking), socioeconomic factors (for example, family instability and economic performance), and built environment characteristics (for example, access to mental health services and street intersection density). Our framework refines previous evidence (Breslau et al., 2014; Stier et al., 2021) and provides new insights into how urban living shapes depression, offering policy relevant targets for intervention.

2. Method

2.1. Datasets

From the Centers for Disease Control and Prevention (Centers for Disease Control and Prevention, 2024), we obtained 2022 county-level data (1838 counties) on the prevalence of depression (*DEP*), prevalence of excessive drinking (*ED*), and number of adults with disabilities (*DIS*). *DEP* is defined as the number of adults who have ever been diagnosed with any form of depression disorders by any health professional, *ED* as the number of adults who self-reported having more than five drinks (men) or more than four drinks (women) on more than one occasion during the previous 30 days, and *DIS* as the number of adults with at least one of the following disability types: hearing, vision, cognition, mobility, self-care, and independent living. We grouped the 1838 counties into 826 Metropolitan and Micropolitan Statistical Areas (referred to as cities) using the Core Based Statistical Areas to Federal Information Processing Standard County Crosswalk file (National Bureau of Economic Research, 2024). For these 826 cities, we assembled complementary data from multiple sources. Access to mental health providers (*MHP*) and number of children who live in single-parent households (*SP*) were obtained from the 2022 County Health Rankings National Findings Report (University of Wisconsin Population Health Institute, 2022). They were defined as the number of providers located within a city and the number of individuals under 18 living with only one caregiving parent, respectively. Annual Gross Domestic Product (*GDP*) in 2022 was taken from the U.S. Bureau of Economic Analysis (U.S. Bureau of Economic Analysis, 2022). Finally, the number of pedestrian-oriented street intersections (*INT*) was collected from the U.S. Environmental Protection Agency Smart Location Database (U.S. Environmental Protection Agency, 2021), where more street intersections support more frequent walking trips and overall walkability. We obtained city boundaries from the U.S. Census Bureau (U.S. Census Bureau, 2024).

2.2. Urban scaling

We conducted an urban scaling analysis for each feature in Table 1 across 826 cities to examine how urban indicators vary with population size (Eppig et al., 2007). To control for spatial autocorrelation in urban scaling, we used an eigenvector spatial filter (ESF) (Griffith & Chun, 2014),

$$Y_i = C N_i^\beta \exp \left(\sum_{r \in S} \gamma_r v_{ir} + \zeta_i \right), \quad (1)$$

Table 1

City level dataset. List of variables used in the current study, together with their acronyms, descriptions, and data sources.

Variable	Description	Source
<i>DEP</i>	Number of adults who have ever been diagnosed with any form of depression disorders	CDC PLACES (Centers for Disease Control and Prevention, 2024)
<i>ED</i>	Number of adults who self-reported having more than five drinks (men) or more than four drinks (women) on more than one occasion during the previous 30 days	CDC PLACES (Centers for Disease Control and Prevention, 2024)
<i>DIS</i>	Number of adults who have at least one of the following disability types: hearing, vision, cognition, mobility, self-care, and independent living	CDC PLACES (Centers for Disease Control and Prevention, 2024)
<i>SP</i>	Number of children who live in a household headed by a single parent	County Health Rankings (University of Wisconsin Population Health Institute, 2022)
<i>MHP</i>	Number of mental health providers	County Health Rankings (University of Wisconsin Population Health Institute, 2022)
<i>GDP</i>	Gross domestic product	U.S. BEA (U.S. Bureau of Economic Analysis, 2022)
<i>INT</i>	Number of pedestrian-oriented street intersections	Smart Location Database (U.S. Environmental Protection Agency, 2021)

where i identifies the generic city, Y_i the urban variable of interest for city i , N_i the population size of city i , β is the scaling exponent, v_{ir} the component of the r -th eigenvector obtained from the spatial weights matrix for city i , S the selected subset of eigenvectors, and ζ_i the residuals of the scaling after adjusting spatial dependence for city i . The slope of the fitted linear model represents the scaling exponent β , where $\beta = 1$, $\beta > 1$, and $\beta < 1$ indicate linear, superlinear, and sublinear scaling, respectively. The spatial weights matrix is constructed as a k -nearest-neighbor graph based on inter-city distances. For each candidate k (ranging from 5 to 15), we fitted the ESF scaling model in Eq. (1) and selected the subset of eigenvectors S through a forward selection procedure that minimizes spatial autocorrelation in the residuals (ζ_i 's) (Griffith & Peres-Neto, 2006) (see Supplementary Information). We then determined the optimal k that yields the lowest cross-validation root-mean-square error across fitted models (see Supplementary Information). ζ_i , defined as features-adjusted metropolitan indicator (FsAMI) (Gan et al., 2025), corresponds to the specific residual of city i that is computed from OLS estimation `statsmodels` (v0.14.2) of the logarithmically transformed function of Eq. (1). Spatial autocorrelation is assessed by Moran's I using the Python library `PySAL` (v23.07) (Rey & Anselin, 2009).

2.3. Causal discovery

Given that FsAMIs allow one to remove spurious dependencies from multiple confounding features, we used them as input to infer associations between urban variables. We implemented the Peter-Clark (PC) algorithm on the FsAMIs for seven variables: prevalence of adult depression (*DEP*), prevalence of excessive drinking (*ED*), number of adults with disabilities (*DIS*), number of children who live in single-parent households (*SP*), access to mental health providers (*MHP*), GDP (*GDP*), and number of street intersections (*INT*). Here, we employed Spearman's rank correlation for independence testing to reflect the monotonic nature of the dataset. Since the PC algorithm assumes no multicollinearity among variables (Kalisch & Bühlmann, 2007), we computed variance inflation factors (VIFs) of the FsAMIs of the seven urban variables using `statsmodels` (v0.14.2) to quantify the extent to which their variance is inflated by correlations with other variables (see Supplementary Information).

We applied the PC algorithm based on the `gCastle` (v1.0.3) (Wang et al., 2022) at a significance level of $\alpha = 0.05$, and computed the Spearman's rank correlation between variables using `SciPy` (v1.16.1). For conditional independence tests, Spearman's rank correlations were calculated between the residuals of both variables after regressing them on the conditioning set. The PC algorithm is order-dependent because

sequential conditional independence tests let early decisions constrain the subsequent search space (Spirtes et al., 2000). Therefore, we ran the algorithm across all $P(7, 7) = 5,040$ permutations of input variables and retained only those edge orientations that appeared in more than half of the runs to ensure the robustness of the inferred causal structure.

3. Results

3.1. Urban scaling analysis

Urban scaling for the seven urban features is presented in Fig. 1. Depression prevalence (*DEP*) scales sublinearly with city population ($\beta = 0.994$ with 95% confidence interval, CI, [0.990, 0.998]), implying that depression prevalence per capita decreases with the population size, see Fig. 1(a). Similarly, the number of adults with disability (*DIS*) and street intersections (*INT*) scale sublinearly (*DIS*: $\beta = 0.959$ with 95% CI [0.954, 0.964]; *INT*: $\beta = 0.896$ with 95% CI [0.875, 0.917]), see Fig. 1(c) and (g), respectively. Conversely, the relationship between the prevalence of excessive drinking and population size is superlinear ($\beta = 1.023$ with 95% CI [1.018, 1.028]). Likewise, the number of children who live in single-parent households (*SP*), access to mental health providers (*MHP*), and GDP, scale superlinearly with population size (*SP*: $\beta = 1.028$ with 95% CI [1.014, 1.043]; *MHP*: $\beta = 1.151$ with 95% CI [1.123, 1.180]; *GDP*: $\beta = 1.081$ with 95% CI [1.066, 1.097]), see Fig. 1(d), (e), and (f), respectively.

The joint distribution of FsAMIs further captures how pairwise variables are associated across cities, see Fig. 2 for an illustration. As depicted in Fig. 3, depression shows weak positive associations with excessive drinking, disability, and mental health providers (*ED*: Spearman's rank correlation coefficient $\rho = 0.082$, $p = 0.018$, *DIS*: $\rho = 0.254$, $p < 0.001$, and *MHP*: $\rho = 0.085$, $p = 0.015$, respectively), but negatively with GDP and street intersections (*GDP*: $\rho = -0.188$, $p < 0.001$, and *INT*: $\rho = -0.082$, $p = 0.018$). GDP also shows negative associations with disability and children in single-parent households (*DIS*: $\rho = -0.273$, $p < 0.001$, and *SP*: $\rho = -0.069$, $p = 0.046$), while positive with excessive drinking and street intersections (*ED*: $\rho = 0.122$, $p < 0.001$, and *INT*: $\rho = 0.150$, $p < 0.001$). Additionally, excessive drinking shows negative associations with disability and children in single-parent households (*ED*: $\rho = -0.408$, $p < 0.001$, and *SP*: $\rho = -0.291$, $p < 0.001$). Meanwhile, disability exhibits strong positive associations with children in single-parent households ($\rho = 0.465$, $p < 0.001$).

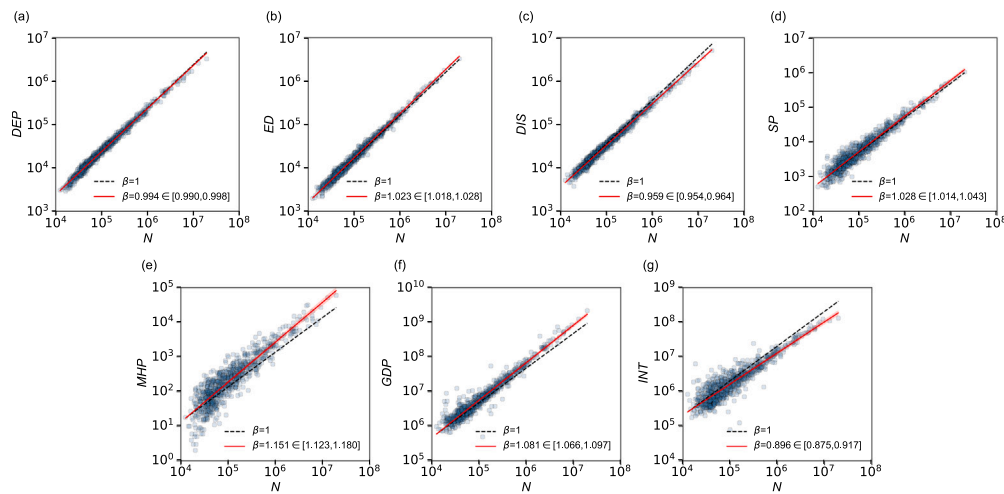


Fig. 1. Urban scaling for 826 cities in the United States. (a) Number of adults who have ever been diagnosed with depression in 2022, (b) number of adults who self-reported excessive drinking in 2022, (c) number of adults who reported at least one of disability types in 2022, (d) number of children who live in single-parent households in 2022, (e) number of mental health providers in 2022, (f) GDP in 2022, and (g) number of pedestrian-oriented street intersections in 2021 as functions of the population size. The dashed line in each plot represents linear scaling, and the shaded area shows the confidence interval.

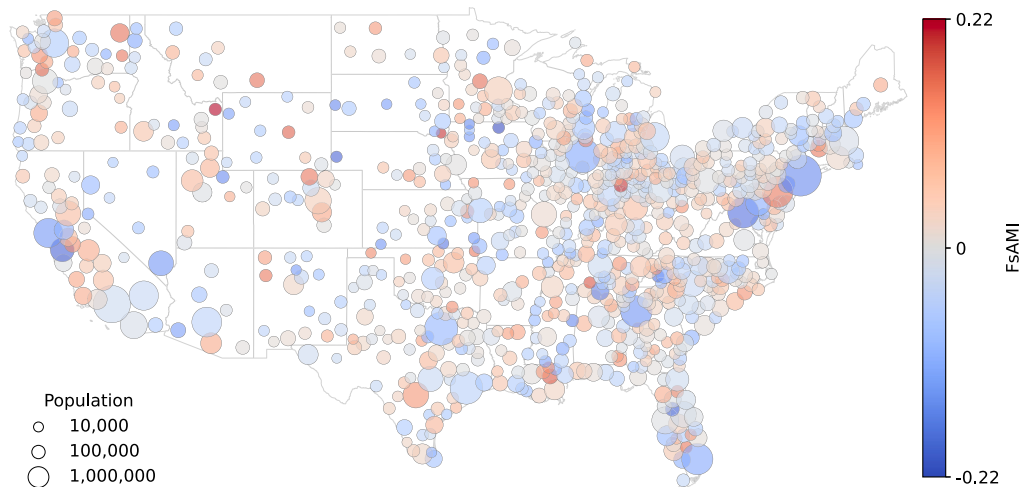


Fig. 2. Map of features-adjusted metropolitan indicators (FsAMIs) for depression prevalence across 826 cities in the U.S. (except for Hawaii and Alaska). Each dot represents a city, and the dot size indicates its population size. The color of each dot represents the FsAMI of number of depression for each city, with red (blue) dots indicating positive (negative) values. US Census shape files (U.S. Census Bureau, 2020) were used to create this map.

3.2. Causal associations

The VIFs of all variables are less than 5, suggesting weak to no collinearity among these variables (*DEP*: VIF = 1.185; *ED*: VIF = 1.295; *DIS*: VIF = 1.761; *SP*: VIF = 1.396; *MHP*: VIF = 1.028; *GDP*: VIF = 1.134; and *INT*: VIF = 1.034). We utilized FsAMIs as the input to a causal analysis that leverages pairwise and partial correlations to identify associations and orientations between the variables under Pearl's framework (Pearl, 2009).

The observed DAG depicted in Fig. 4 suggests that increased prevalence of excessive drinking is associated with a higher number of adults suffering from depression in a city. We observed that a higher prevalence of disability is associated with an increased prevalence of depression and a lower prevalence of excessive drinking. Access to mental health providers is positively associated with the prevalence of depression. Notably, we observed that a larger number of street intersections and higher GDP are correlated with a lower prevalence of depression, while the number of street intersections is positively associated with the GDP. Meanwhile, the prevalence of disability is negatively associated with GDP, and no causal relationship is detected in

either direction. The increased prevalence of single-parent households is related to increased prevalence of disability and decreased prevalence of excessive drinking.

4. Discussion

Compared to rural environments, living in cities has been proposed to increase the likelihood to develop depressive disorders in their dwellers (C. Xu et al., 2023). Yet, and in line with other scholars (Stier et al., 2021), our findings support an opposite perspective; we find that larger cities tend to exhibit lower per capita rates of depression. This may suggest that while larger cities may pose some pathogenic features, at the same time, they may also offer protective features capable of counteracting urban-related vulnerability to depression. To uncover the underlying dynamics, we aimed to investigate the respective roles of several variables previously implicated in the onset of depressive disorder, either as vulnerability or protective factors. We thus selected validated proxies of excessive drinking, disability, and marital dissolution as vulnerability factors, and road intersections, availability of mental healthcare providers, and economic wealth as protective

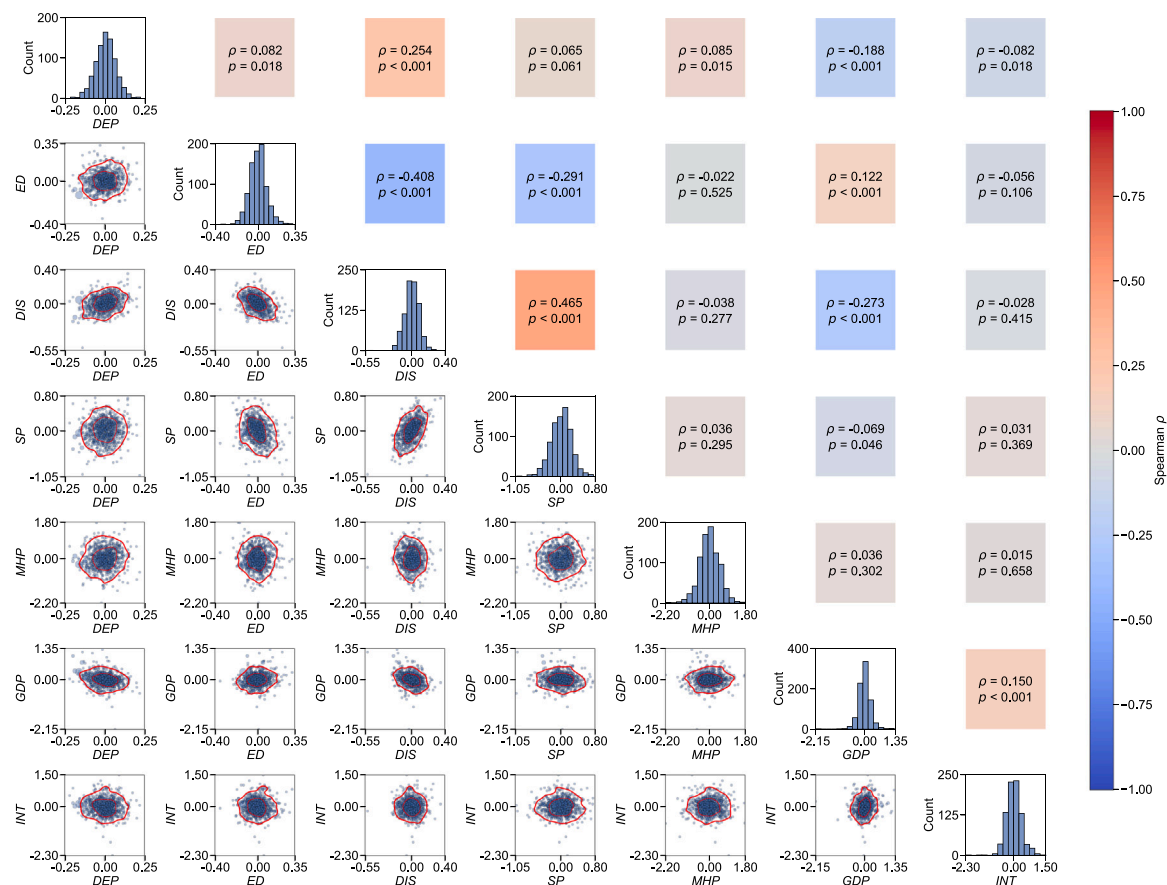


Fig. 3. Distribution of features-adjusted metropolitan indicators (FsAMIs) across 826 cities in the United States. Scatter plots show marginal and pairwise joint distributions of FsAMIs of the six chosen variables: prevalence of adult depression (DEP), prevalence of excessive drinking (ED), number of adults with disabilities (DIS), number of children who live in single-parent households (SP), access to mental health providers (MHP), GDP (GDP), and number of street intersections (INT). Each dot represents a city, with its size proportional to the population. Larger dots indicate larger populations. Histograms display the marginal distributions of the FsAMIs, annotate Spearman's rank correlation (ρ) between each feature's FsAMI and city population, with corresponding p values. Upper-triangle panels report pairwise Spearman's rank correlation (ρ) with p values.

factors. We then explored the scaling behaviors of these variables and investigated causal associations among them. Our approach aims to offer insight into the mechanisms potentially moderating depression prevalence across different cities, and identifying population-level variables apt to large-scale interventions in the pursuit of individual mental well-being.

The sublinear scaling of depression prevalence with population size aligns with the scaling result reported by Stier et al. (2021). However, their scaling analysis did not explicitly account for spatial interactions among cities, which may bias the estimation of scaling exponents. Although Huth et al. (2022) proposed that the sublinear scaling relationship between population size and depression is sensitive to the definition of city boundaries, Stier et al. (2022) argued that cities are defined by aggregated socioeconomic networks rather than different radial buffers around city centers. To address the spatial delineation issue, we incorporated spatial filtering into our urban scaling models and demonstrated that the sublinear scaling of depression prevalence remains robust, even after controlling for spatial dependencies. This clarifies that the observed pattern is robust and not an artifact of spatial proximity.

Our urban scaling results also revealed the critical role of urbanization in shaping human well-being. Similar to depression prevalence, the number of adults with disabilities scales sublinearly, suggesting that health burdens grow more slowly than population size. This finding aligns with previous evidence reporting that persons with disabilities experience more barriers in accessing health resources in smaller cities (Hamilton et al., 2022). Beyond these health-related outcomes,

our results also uncover the sublinear relationship between street intersections and population size, in agreement with prior findings about the economy of scale of shared infrastructure (Ribeiro et al., 2017). Economic and professional indicators follow superlinear trends, instead. Both GDP and access to mental health providers rise faster than city size, illustrating the concentration of wealth and specialized services in large cities, which are also observed in broader analyses of urban inequality in the U.S. (Nijman & Wei, 2020). Superlinear trends also characterize social risk factors. Excessive drinking exhibits a superlinear pattern, reflecting the intensification of socially mediated behaviors in dense urban environments where nightlife and social gatherings are more prevalent (McCulley et al., 2022; Mullachery et al., 2022). Number of children who live in single-parent households also scales superlinearly with city size, potentially capturing the heightened socioeconomic pressures and demographic vulnerabilities in large cities (McLanahan & Percheski, 2008). Together, these findings highlight that the sublinear scaling of depression is accompanied by mixed patterns of potential urban determinants, including sublinear growth in number of adults with disabilities and street intersections, alongside superlinear growth in excessive drinking, single-parent households, economic prosperity, and mental health services.

Our causal framework further echoes this duality. Health behavioral factors emerged as vital pathways linking city life to mental health, with high levels of disability and excessive drinking worsening depression prevalence. Prior studies (Dalton et al., 2003; Na & Yang, 2022; Senra et al., 2015) have reported a similar association between depression and disability, suggesting that persons with disabilities, such

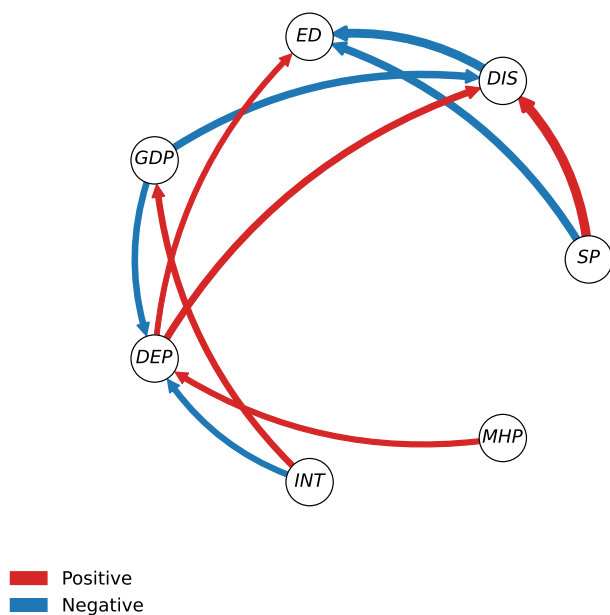


Fig. 4. Causal graph among seven urban features, inferred from data across 826 U.S. cities. Directed Acyclic graph among six urban features, estimated by Spearman's rank correlation with a significance level of 0.05: prevalence of depression (DEP), prevalence of excessive drinking (ED), number of adults with disabilities (DIS), number of children who live in single-parent households (SP), access to mental health providers (MHP), GDP (GDP), and number of street intersections (INT). Red arrows indicate positive association, while blue arrows indicate negative association. Undirected edges indicate interactions that cannot be oriented. The thickness of edges identifies the pairwise Spearman's rank correlation coefficient; a thicker edge represents a higher value of correlation.

as visual or hearing impairments, or mobility and self-care limitations, tend to have lower health-related quality of life and heightened vulnerability to depressive disorders. Similarly, excessive drinking represents a major vulnerability factor in depression, whereby it interferes with neurotransmitter systems involved in mood regulation and reward sensitivity. In fact, alcohol has been reported to modulate the brain concentrations of serotonin (Coppen, 1967) (a monoamine traditionally implicated in the etiology and treatment of depression), the GABAergic transmission in the amygdala (Chen et al., 2025) (interfering with emotional regulation), and the dopaminergic transmission in the ventral tegmental area (Brodie et al., 1999) (contributing to an alteration in reward sensitivity).

Conversely, higher street intersection density is associated with lower depression prevalence. When considering candidate neuroanatomical mediators of depression, our findings align intriguingly with studies linking intersection density to hippocampal volume, given the well-established role of the hippocampus in the pathogenesis and course of depression (Brown et al., 2014). The influence of street intersections on hippocampal volume may occur via both the promotion of physical activity and the increased cognitive load associated with spatial navigation. With respect to physical activity, Khalil (2025) systematically reviewed the studies examining hippocampal volume and walking and observed a positive association. At the same time, Shin et al. (2024) investigated the association between environmental complexity and hippocampal brain volume in 660 adult volunteers and observed that living in more complex environments resulted in increased volumes in hippocampal subregions involved in spatial navigation. Last, Cerin et al. (2017) reported that “walkable” neighborhoods were associated with better neuroimaging outcomes, and that these effects were minimally attributable to physical activity.

Ultimately, these studies lend support to the hypothesis that street intersections may influence brain areas involved in mood regulation but that this association may extend beyond physical activity. Economic hubs appear to counteract vulnerability factors. Higher GDP in larger cities, indicating greater aggregate income and overall wealth, is associated with lower depression and disability prevalence. This aligns with the existing literature (Ettman et al., 2023; Thomson et al., 2022), suggesting that economic prosperity may support healthier living conditions. Intriguingly, high provision of mental health professionals in large cities is positively associated with depression prevalence. Although this evidence may appear counterintuitive, we offer that it primarily reflects better access to diagnosis (Santos et al., 2024). Although no direct cause-and-effect links are detected, we observed that a higher prevalence of single-parent households indirectly increases the proneness to depression, moderated by disability. The elevated threat of single parenthood aligns with the reported evidence of marital dissolution as a potential risk factor in depression (Zhai et al., 2024). These causal pathways underline how city life contributes to and buffers against depression through intertwined behavioral, socioeconomic, and infrastructural mechanisms.

The study is not free of limitations. First, the causal discovery framework implemented through the PC algorithm is based on several strong assumptions, including the absence of hidden confounders, the causal Markov condition, and acyclicity. Violations of these assumptions may challenge the inferred causal structure. Second, the cross-sectional design limits our ability to fully establish temporal precedence, and future longitudinal or quasi-experimental studies are needed to validate the robustness of the findings. Finally, although the integration of urban scaling and causal discovery offers novel insights, the interpretation of specific mechanisms should be approached with caution, as observational data cannot fully rule out residual confounding. In addition, this study does not account for natural environmental factors, such as green space availability, air quality, or climate changes, which may also influence mental health outcomes, representing an important direction for future research (Bezold et al., 2018).

Despite these limitations, our work offers a novel lens to explore the relationship between depression prevalence and city living, shaped by a combination of behavioral, socioeconomic, and environmental factors. Our findings corroborate the claim that depression is less prevalent in large cities, where both vulnerable determinants, including disabilities and excessive drinking, and protective determinants, including economic prosperity and walkability, provide risks and protections against depression. Our findings suggest that small cities are more vulnerable, calling for local municipalities and state officials to promote economic development, healthier lifestyles, and equitable mental health care, supporting a protective and healthy system for the residents.

CRediT authorship contribution statement

Junyi Li: Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation. **Tian Gan:** Writing – original draft, Validation, Methodology, Conceptualization. **Shenghan Lyu:** Data curation. **Hongying Wu:** Data curation. **Simone Macri:** Writing – review & editing, Supervision, Resources, Conceptualization. **Maurizio Porfiri:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization.

Code availability

Scripts that are allowed to be shared are available on [The Dynamical Systems Laboratory's Github](#).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary information

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.cities.2026.107215>.

Data availability

Datasets that are allowed to be shared are available on The Dynamical Systems Laboratory's Github.

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